Leonel Hernandez

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Science — *May 2029*

Experience

Lowes Sale Associate — Aug. 2025 – Present
Lowes — *Kissimmee, FL*

- Used teamwork, clear communication, and task prioritization to coordinate department work and complete assigned tasks on schedule in a busy working environment
- Prioritized department tasks and managed time across assignments, ensuring work was completed by deadlines

Kumon Tutor/Software Assistant — Nov. 2024 – Jul. 2025
Kumon — *Kissimmee, FL*

- Developed a functional **web application portal** in collaboration with the business owner, streamlining student application submissions and administrative workflows.
- Automated student class reminders through integration with the **Twilio API**, increasing class attendance by over **25%**.
- Maintained production web application and SMS functionality, troubleshooting technical issues to support reliable daily operations.

Projects

Guiding Hands | *Python, Google Media Pipeline*

- Engineered a real-time, gesture-controlled stroke rehabilitation game using **Python**, **MediaPipe**, and **NVIDIA Jetson Orin Nano**, translating hand-tracking data into interactive gameplay.
- Accelerated **MediaPipe** hand-tracking inference by **9.46×** using **GPU acceleration** on the **NVIDIA Jetson Orin Nano**, benchmarked against CPU inference on identical video input.
- Collaborated with a **4-person engineering team** to integrate gesture recognition, real-time tracking, and game mechanics into a functional rehabilitation prototype.

mySPIM Processor | *C/C++*

- Developed a C-based mySPIM architectural simulator modeling a 32-bit MIPS processor instruction set executing a total of **14 ALU, memory, and branch instructions**.
- Constructed core logic and control unit logic while managing 32 general purpose registers & instruction decoding.
- Designed ALU to handle dynamic binary operations, bit buffering, sign extension and status flags (Zero, Overflow)
- Utilized core **C low level memory management** (pointers, bitwise operators, data structures) in order to simulate main memory and control registers

PM/0 Machine | *C/C++*

- Implemented a stack-based **PM/0 virtual machine** in **C**, simulating instruction execution through a **3-register architecture** consisting of the Program Counter, Base Pointer, and Stack Pointer.
- Designed runtime stack tracing and activation-record management to support procedure calls, control flow, and nested execution contexts.

Involvement

Knight Hacks Mentee — September 2026
Knight Hacks Mentorship Program

- Joined the Knight Hacks Mentorship Program as a mentee under Sebastian Noel and attended various Knight Hacks workshops and hackathons in order to improve technical skills and develop technical conections.

Technical Skills

Languages: Java, Python, C, C++, HTML, CSS
Technologies: Git, GitHub, Twilio API, Google Media Pipeline
Concepts: Data Structures, Object-Oriented Programming, Pointers, Memory Management, Bitwise Operations, Basic Computer Architecture, Software Development, Linear Algebra, Algorithms, Statistics